

ASSIGNMENT

AI-Assisted Literature Review: A Comparative Evaluation of Research Tools

Course: AI Tools for Research
Individual Assignment

Topic ID: T-G11

Assigned Topic: Single-LLM Versus Multi-Agent Debate Systems for Scientific Hypothesis Generation

1. Purpose of the Assignment

This assignment focuses specifically on producing a literature review with the assistance of multiple research-oriented AI tools and evaluating how the tools differ in the literature they identify and the review they help produce. It is a synthesis-and-comparison activity, not another citation-audit or citation-mapping exercise.

Earlier laboratory work has already addressed citation verification, citation authenticity, systematic reference sampling, claim–citation support, and citation mapping. Those activities should not be repeated here. In this assignment, students will use different tools to generate literature-review inputs and then compare the tools on the basis of the usefulness, coverage, relevance and quality of the literature-review output.

2. Assignment Objective

Using the same assigned research topic, students will independently explore the literature with four research tools, use the information obtained to produce a literature review, and then compare the tools critically.

The assignment therefore has two connected outputs:

- 1. A literature review/research paper produced with the assistance of the selected tools.
- 2. A comparative evaluation explaining which tools were more useful, what each tool contributed, and which workflow the student recommends.

3. Tools to Be Used

Tool	Role in This Assignment	Primary Contribution
ResearchRabbit	Literature discovery	Related, earlier and later research; authors and literature connections
Litmaps	Literature exploration	Evolution of literature, related studies and chronological development
Semantic Scholar	Scholarly search	Relevant, influential and recent research discovery
Elicit	Literature synthesis support	Structured extraction and comparison of research questions, methods, findings and limitations

Students must use all four tools. The objective is not to prove that one tool is universally superior, but to determine which tool is most useful for particular stages of a literature review.

4. Part A — Understand the Assigned Topic

Before using the tools, define the scope of your literature review.

Item	Student Response
Topic ID	T-G11
Assigned Topic	Single-LLM Versus Multi-Agent Debate Systems for Scientific Hypothesis Generation
Main Research Question	How do single-LLM and multi-agent debate approaches compare for generating scientifically novel, valid, and useful hypotheses?
Scope of Review	Studies of LLMs in generating scientific hypothesis/research idea generation through various approaches such as single LLM approach, iterative approach, multi-agent approach, agent specialization, literature grounding, assessment, and experimentation.
Key concepts to be covered	Generation of single-LLM hypotheses; iteration and self-criticism; debate among multiple agents; diversity and independent thinking; scientific novelty and validity; information search and grounding; evaluation techniques; computational expense; experimentation.

5. Part B — Explore the Literature Using Four Tools

Use the assigned topic and suitable search terms in each tool. Students may use different queries where the tool's function requires them, but must record the important prompts/search instructions used.

B1. ResearchRabbit

- Use the topic and/or suitable seed papers to explore related literature.
- Identify papers that appear particularly relevant to the review.
- Explore earlier and later research where useful.
- Use the resulting literature to identify major themes or research directions.
- Record what ResearchRabbit contributed to the final review.

B2. Litmaps

- Explore the literature surrounding the assigned topic.
- Use the tool to understand how research on the topic has developed.
- Identify relevant studies and emerging areas.
- Record literature discovered through Litmaps that was not already identified through ResearchRabbit.

B3. Semantic Scholar

- Search the assigned topic using appropriate queries.
- Identify influential and recent papers.
- Explore related literature.
- Select papers that contribute directly to the final review.
- Record the contribution of Semantic Scholar to the final literature set.

B4. Elicit

- Use Elicit to explore literature relevant to the research question.
- Use it to compare papers on research problem, methodology, findings and limitations.
- Identify themes, methodological patterns and unresolved issues.
- Use the extracted information as an aid to synthesis, not as a replacement for reading the papers.

6. Part C — Create the Literature Set for the Review

After exploring all four tools, create a final set of papers that will form the evidence base for the literature review. This is a selection-and-synthesis task rather than a citation-audit exercise.

Recommended minimum: 20 scholarly papers, with a reasonable balance of foundational, influential and recent work.

Paper ID	Paper / Authors	Year	Discovered Through	Why Important for the Review	Included?
P01	Accelerating scientific discovery with Co-Scientist — Gottweis et al.	2026	ResearchRabbit, Litmaps, Elicit	Major multi-agent scientific-discovery system using generate–debate–evolve and experimental validation.	Yes
P02	Can LLMs Generate Novel Research Ideas? — Si et al.	2025	ResearchRabbit, Litmaps	Human evaluation of LLM-generated research ideas, especially novelty, feasibility and diversity.	Yes
P03	The AI Scientist: Towards Fully Automated Open-Ended Scientific Discovery — Lu et al.	2024	ResearchRabbit, Litmaps	Important single-agent/autonomous research baseline covering ideation through experiments and writing.	Yes
P04	ResearchAgent: Iterative Research Idea Generation over Scientific Literature with Large Language Models — Baek et al.	2024	ResearchRabbit, Elicit	Directly relevant to iterative, literature-grounded idea generation and review.	Yes
P05	SciMON: Scientific Inspiration Machines Optimized for Novelty — Wang et al.	2024	ResearchRabbit, Elicit	Directly studies novelty optimization in literature-grounded LLM idea generation.	Yes
P06	Learning to Generate Novel Scientific Directions with Contextualized Literature-based Discovery — Wang et al.	2023	ResearchRabbit	Connects literature-based discovery with generation of novel scientific directions.	No
P07	Agent Laboratory: Using LLM Agents as Research Assistants — Schmidgall et al.	2025	ResearchRabbit	Extends LLMs toward autonomous research workflows and experimentation.	No
P08	Large Language Models are Zero Shot Hypothesis Proposers — Qi et al.	2023	ResearchRabbit, Semantic Scholar, Elicit	Foundational direct test of single-LLM scientific hypothesis generation.	Yes
P09	SciAgents: Automating Scientific	2024	ResearchRabbit	Relevant multi-agent architecture combining	No

	Discovery through Multi-Agent Intelligent Graph Reasoning			scientific knowledge and reasoning.	
P10	Improving Scientific Hypothesis Generation with Knowledge Grounded Large Language Models — Xiong et al.	2024	ResearchRabbit, Semantic Scholar	Important for understanding the role of external knowledge and grounding.	Yes
P11	Undiscovered Public Knowledge — Swanson	1986	Litmaps	Foundational literature-based-discovery concept behind connecting disconnected knowledge.	No
P12	Reflexion: An Autonomous Agent with Dynamic Memory and Self-Reflection — Shinn et al.	2023	Litmaps	Key single-agent self-reflection approach for comparison with multi-agent systems.	No
P13	Improving Factuality and Reasoning in Language Models through Multiagent Debate — Du et al.	2024	Litmaps	Foundational multi-agent debate paper and direct basis for the debate paradigm.	Yes
P14	Large Language Models for Automated Open-domain Scientific Hypotheses Discovery — Yang et al.	2024	Litmaps	Direct open-domain scientific hypothesis-generation system.	Yes
P15	Hypothesis Generation with Large Language Models — Zhou et al.	2024	Litmaps, Semantic Scholar, Elicit	Direct single-LLM/data-driven hypothesis-generation baseline with iterative refinement.	Yes
P16	The Virtual Lab: AI Agents Design New SARS-CoV-2 Nanobodies with Experimental Validation	2024	Litmaps	Multi-agent scientific discovery with actual laboratory validation.	Yes
P17	CONSENSAGENT: Towards Efficient and Effective Consensus in Multi-Agent LLM Interactions Through Sycophancy Mitigation — Pitre et al.	2025	Litmaps	Important for multi-agent failure modes such as sycophancy and unreliable consensus.	Yes
P18	Towards End-to-End Automation of AI Research — Yamada et al.	2026	Litmaps	Latest evidence on the capabilities and limitations of autonomous AI research.	Yes

P19	A Survey of AI Scientists: Surveying the Automatic Scientists and Research	2025	Litmaps	Useful overview for organizing the broader autonomous-scientist literature.	No
P20	A Survey on Hypothesis Generation for Scientific Discovery in the Era of Large Language Models	2025	Semantic Scholar	Direct overview of the hypothesis-generation research landscape.	No
P21	IdeaBench: Benchmarking Large Language Models for Research Idea Generation — Guo et al.	2025	Semantic Scholar	Important benchmark for measuring research-idea quality.	Yes
P22	Toward Reliable Biomedical Hypothesis Generation: Evaluating Truthfulness and Hallucination in Large Language Models — Xiong et al.	2025	Semantic Scholar	Addresses truthfulness and hallucination, important limitations of LLM-generated hypotheses.	Yes
P23	Many Heads Are Better Than One: Improved Scientific Idea Generation by a LLM-Based Multi-Agent System — Su et al.	2025	Semantic Scholar, Elicit	Directly compares multi-agent scientific ideation with single-agent baselines.	Yes
P24	Do LLMs Know a Good Hypothesis When They See One? Logit-Based Energy Scoring Outperforms Prompted LLM-as-Judge for Scientific Hypothesis Ranking — Rajwal et al.	2026	Semantic Scholar	Important for the problem of evaluating and ranking generated hypotheses.	Yes
P25	Can Large Language Models as Biomedical Hypothesis Generators: A Comprehensive Evaluation — Qi et al.	2024	Elicit	Broad evaluation of novelty, correctness and quality of LLM-generated biomedical hypotheses.	Yes
P26	Can ChatGPT be used to generate scientific hypotheses? — Park et al.	2023	Semantic Scholar, Elicit	Early direct study of LLM scientific hypothesis generation, including error and usefulness.	No
P27	Two Heads Are Better Than One: A	2024	Elicit	Early direct multi-agent vs single-agent	Yes

	Multi-Agent System Has the Potential to Improve Scientific Idea Generation — Su et al.			scientific-idea comparison.	
P28	Exploring Design of Multi-Agent LLM Dialogues for Research Ideation — Ueda et al.	2025	Elicit	Directly studies agent number, role diversity and dialogue depth.	Yes
P29	HypoBench: Towards Systematic and Principled Benchmarking for Hypothesis Generation — Liu et al.	2025	Elicit	Evaluates practical utility, generalizability and hypothesis discovery across many tasks.	Yes
P30	BioDisco: Multi-agent hypothesis generation with dual-mode evidence, iterative feedback and temporal evaluation — Ke et al.	2025	Elicit	Strong recent example combining multi-agent generation, evidence grounding and temporal evaluation.	Yes

The attached spreadsheet has a list of 72 papers, out of which only 30 were included in this table and 22 were selected for writing the final paper.

7. Part D — Generate the Literature Review

Using the literature identified with the four tools, generate a coherent literature review/research paper on the assigned topic. The review must be organised around themes, approaches, findings, limitations and research gaps rather than paper-by-paper summaries.

Required Structure

3. Title
4. Abstract
5. Keywords
6. Introduction
7. Background / Conceptual Foundation
8. Thematic Literature Review
9. Comparison of Existing Approaches
10. Research Trends / Developments
11. Challenges and Limitations
12. Research Gaps
13. Future Research Directions
14. Conclusion
15. References

Recommended length: 8–12 pages, excluding references and appendices.

8. Part E — Tool-by-Tool Contribution to the Review

After completing the literature review, identify how each tool contributed to the final paper.

Tool	Papers Contributed	Papers Unique to Tool	Most Useful Contribution	Limitations Observed
ResearchRabbit	7	0	Helped visualize citation relationships and discover papers connected to the initial seed articles. Its citation-network view was useful for identifying influential papers, related studies, and earlier foundational work.	Most papers overlapped with results from other tools. The quality of recommendations depended heavily on the initial seed papers, and some results were only indirectly related to scientific hypothesis generation.
Litmaps	9	5	Helped expand the literature through citation links and identify research clusters around LLM-based scientific discovery, multi-agent systems, and automated research. It was especially useful for tracing how newer papers developed from earlier studies.	Citation-based discovery sometimes produced papers that were technically connected but not directly relevant to the comparison between single-LLM and multi-agent systems. The map could also become difficult to interpret when many papers were added.
Semantic Scholar	7	3	Provided a fast way to search for relevant papers using keywords, authors, citations, and related-paper recommendations. Its citation counts, paper summaries, and topic-based search helped prioritize influential and recent studies.	Search results sometimes favored highly cited general LLM or agent papers rather than papers specifically focused on scientific hypothesis generation. Some papers were duplicated across other tools, and citation counts did not necessarily indicate direct relevance to the research question.
Elicit	11	5	Was the most useful tool for directly comparing papers. Its structured table made it easier to extract research objectives, methods, datasets, findings, and limitations. It helped identify papers discussing agent number, role diversity, dialogue depth, novelty, validity, and practical usefulness.	The quality of the extracted information depended on the paper's availability and clarity. Some summaries were incomplete or required manual verification. Elicit also sometimes grouped papers with related themes even when their experimental settings were not directly comparable.

9. Part F — Comparative Analysis of the Tools

Rate each tool from 1 to 5 based on your actual experience during this assignment.

Criterion	ResearchRabbit	Litmaps	Semantic Scholar	Elicit
Ease of use	5	4	5	4
Relevance of literature discovered	4	4	4	5
Coverage of the topic	3	4	5	4
Discovery of recent research	4	4	5	4
Discovery of foundational research	5	5	4	3
Discovery of less obvious/related research	4	5	5	4
Support for identifying themes	4	4	3	5
Support for comparing papers	3	3	3	5
Support for identifying research gaps	3	4	3	5
Quality of final literature-review input	4	4	4	5
Overall usefulness	4	3	4	5

Rating scale: 1 = Very Poor; 2 = Poor; 3 = Moderate; 4 = Good; 5 = Excellent.

10. Part G — Compare the Literature Produced by the Tools

Answer using evidence from your own work.

Q1. Which tool produced the most relevant literature for your topic?

Elicit produced the most relevant literature because its results were closely connected to scientific hypothesis generation, LLM evaluation, and multi-agent systems. Its structured summaries also made it easier to select papers directly useful for the research question.

Q2. Which tool produced the broadest literature coverage?

Semantic Scholar provided the broadest coverage through keyword searches, related-paper recommendations, citation links, and author searches. It helped locate papers from LLM research, scientific discovery, multi-agent systems, and evaluation benchmarks.

Q3. Which tool was most useful for finding recent research?

Semantic Scholar was the most useful for finding recent research. Its search results and related-paper features helped identify newer work on AI scientists, agentic research systems, scientific debate, and automated discovery.

Q4. Which tool was most useful for finding foundational or influential work?

ResearchRabbit and Litmaps were the most useful for finding foundational and influential work. Their citation-network maps helped trace papers backward to important earlier studies and forward to newer developments.

Q5. Which tool helped most with identifying themes and research directions?

Elicit helped most with identifying themes and research directions. Its paper summaries revealed recurring themes such as novelty, usefulness, agent diversity, debate depth, scientific validation, reliability, and research automation.

Q6. Which tool helped most with comparing methodologies and findings?

Elicit was the most useful for comparing methodologies and findings. Its table-based format made it easier to compare research objectives, experimental methods, datasets, results, and limitations across papers.

Q7. Which tool contributed the most papers that were not discovered through the other tools?

Elicit and Litmaps contributed the most unique papers, with five papers each. Elicit found unique papers through topic-focused searching, while Litmaps discovered unique papers through citation-network expansion.

Q8. Which tool produced the most irrelevant or less useful results?

Semantic Scholar produced the highest number of less directly relevant results. It often returned general papers on LLMs, AI agents, or reasoning that were useful for background but not specifically focused on scientific hypothesis generation.

Q9. Did the use of four tools materially improve the literature review compared with relying on one tool?

Yes, using four tools materially improved the review by increasing coverage, reducing dependence on one search method, and revealing both foundational and recent research. The tools also complemented one another: mapping tools supported discovery, Semantic Scholar supported searching, and Elicit supported comparison and synthesis.

Q10. Which combination of tools would you use for a future literature review and why?

I would use Semantic Scholar, ResearchRabbit, and Elicit. Semantic Scholar would provide broad and recent papers, ResearchRabbit would expand citation networks, and Elicit would organize and compare the final literature set efficiently.

11. Part H — Compare the Final Review Output

If feasible, generate a short literature-review response from each tool using comparable instructions. Then compare the outputs before preparing the final integrated review.

Dimension	ResearchRabbit	Litmaps	Semantic Scholar	Elicit
Organisation of review	The papers were organized mainly through citation maps and collections. A coherent review required additional manual arrangement.	The citation map helped arrange papers chronologically and by research connections, but thematic organization required manual work.	The search results were easy to collect, but the papers needed to be grouped manually into themes for a structured review.	The paper table naturally supported organization by objectives, methods, findings, and limitations, making the review easier to structure.
Coverage of major themes	Covered foundational research, scientific discovery, LLMs, and related agent systems through connected papers.	Covered major themes through citation expansion, especially the development from earlier work to newer agentic systems.	Provided the broadest coverage, including single-LLM systems, multi-agent debate, scientific discovery, and evaluation.	Covered the main themes most directly, particularly hypothesis generation, debate, novelty, usefulness, and evaluation.

Depth of synthesis	Strong for showing relationships between papers, but weaker for comparing detailed findings and methodologies.	Helped show how research developed over time, but did not directly synthesize experimental results.	Provided useful paper information, although deeper synthesis required reading abstracts and papers manually.	Supported the deepest synthesis because papers could be compared using structured information about methods, results, and limitations.
Identification of gaps	Citation gaps and underexplored connections could be noticed from the map, but research gaps were not explicitly stated.	Helped reveal areas with fewer papers or limited citation development, although interpretation was manual.	Related-paper results helped identify missing areas, but research gaps were not directly extracted.	Made it easier to identify gaps such as limited experimental validation, high computational cost, weak novelty evaluation, and reliability concerns.
Usefulness for academic writing	Useful for finding references and understanding citation relationships, but additional reading and writing were necessary.	Useful for building the bibliography and understanding the historical development of the topic.	Useful for finding relevant papers, recent studies, and citation information for supporting academic claims.	Most useful for academic writing because its structured summaries supported direct comparison and paragraph development.
Overall quality	A useful discovery and mapping input, especially for foundational and related papers, but not sufficient alone for a complete review.	A strong citation-based discovery input, particularly for tracing research development and finding connected studies.	A broad and informative source of literature, although some general LLM and agent papers required filtering.	The strongest input for the final review because it combined relevant papers with structured comparative information.

Students should base the comparison on the actual outputs produced, not on general descriptions of the tools.

12. Part I — Prompts Used

Submit the exact prompts/instructions used with the tools. The prompt record is a required part of the assignment. Where a tool primarily uses search queries rather than conversational prompts, record the exact search query.

Prompt ID	Tool	Exact Prompt / Search Query	Purpose	Output Used in Review?
PR01	ResearchRabbit	Single-LLM versus multi-agent debate systems for scientific hypothesis generation	To identify initial seed papers and explore related literature through citation networks.	Yes

PL01	Litmaps	Single-LLM versus multi-agent debate systems for scientific hypothesis generation	To create a literature map and discover connected papers through backward and forward citations.	Yes
PS01	Semantic Scholar	large language models scientific hypothesis generation	To find papers focused on using LLMs for generating scientific hypotheses and research ideas.	Yes
PS02	Semantic Scholar	multi-agent debate LLM scientific discovery hypothesis generation	To find research on multi-agent debate, critique, collaboration, and automated scientific discovery. Results were examined using relevance, recency, and citation-count sorting.	Yes
PS03	Semantic Scholar	AI Scientist automated scientific discovery research agents	To locate recent and influential work on autonomous research agents and end-to-end scientific discovery.	Yes
PE01	Elicit	How do single-LLM systems compare with multi-agent debate systems for scientific hypothesis generation?	To identify papers directly comparing single-LLM and multi-agent approaches and extract their main findings.	Yes
PE02	Elicit	What are the main methods, findings, limitations, and research gaps in LLM-based scientific hypothesis generation and multi-agent debate systems?	To organize papers according to methodologies, results, limitations, evaluation criteria, and research gaps.	Yes

Add additional prompt rows as required. Do not submit only screenshots; provide the actual text of the prompts.

13. Part J — Student's Critical Evaluation

Write 500–700 words giving your own critical assessment of the four tools. Address:

- Which tool was most useful overall, and why?
- Which tool was most useful for a specific stage of literature review?
- What important literature did one tool identify that others did not?
- Where did the tools provide overlapping or repetitive results?
- Where did the tools fail to provide useful literature?
- How much human judgement was required to transform tool outputs into a credible literature review?
- What workflow would you recommend to another student conducting a literature review on the same topic?

Final Evaluation

Elicit was the most useful of the four tools in general for this literature review. The primary benefit of it was that it allowed for paper discovery and comparison and organization of research results. It proved particularly useful for finding papers on scientific hypothesis generation, multi-agent systems, research methods, limitations, and future research gaps. The presentation of papers was structured which facilitated comparison of the objectives, methodology, results and conclusions of the papers. But it was not always enough, as some of the results had to be checked manually, and the extent of coverage was sensitive to the wording of the research question.

Each tool was found to be useful in a different stage of the literature review. Semantic Scholar proved most helpful in the beginning search and expansion phase. It offered wide scope and was useful in finding recent research in a timely manner. The sorting features, especially relevance, recency and citation count, were useful as they showed different kinds of important papers. Sorting by relevance enabled us to locate studies that were closely related, and sorting by recency allowed us to identify more recent advances in multi-agent debate and autonomous scientific discovery. Sorting by citations facilitated the identification of influential and foundational research. The tools that were most helpful were ResearchRabbit and Litmaps, which enabled exploration of citation networks and tracking of ideas. They assisted in the identification of papers and related studies that may not be readily found by keyword searches. Elicit was most helpful in the comparison and synthesis phase.

The tools provided an important contribution in identifying literature on automated scientific discovery and AI co-scientists. The review extended to other systems like the AI Scientist, AI co-scientist, Coscientist, and other scientific-agent systems, which were explored in papers. The review was not limited to the LLM-based hypothesis generation systems; papers on the AI Scientist, AI co-scientist, Coscientist, and other scientific-agent systems were also covered. These works demonstrated the potential for integrating planning, tool utilization, experimentation, critique, and iterative refinement with LLMs. A few of these papers were found more easily by using Semantic Scholar's recent paper search and citation exploration tools than other tools. Elicit, however, proved to be very useful to link these papers to a wider context (novelty, validity, feasibility, agent diversity, depth of debate).

There was considerable overlap between the tools. All four papers on large language models, scientific discovery, multi-agent systems, self-refinement, and automated research were returned. The same foundational papers and highly cited studies were seen again and again, particularly if the same seed topic or keyword was utilized. This repetition helped to identify some papers as being central to the field, but it also led to more duplication of screening. The results of ResearchRabbit and Litmaps were quite comparable, as both heavily depend on citation relationships. Additionally, there is overlap in the coverage of recent LLM and agent research between Semantic Scholar and Elicit.

The tools were also limited. Occasionally, papers were returned that were broadly related to LLM or autonomous agents but not directly related to scientific hypothesis generation by Semantic Scholar. ResearchRabbit and Litmaps might be able to find papers that are historically or conceptually related to the paper at hand, but not necessarily directly relevant to the research question. Elicit sometimes gave incomplete or imperfectly extracted information, which needed to be checked in the original paper. Without human assessment, none of the tools could reliably assess the methodologic soundness, experimental support, or originality of a paper's claims.

A lot of judgment had to be used by humans. I had to filter out duplicates, evaluate relevance, determine if the paper was about foundational work or recent developments, confirm paper information, and determine if a paper was about single-LLM systems, multi-agent debate, or scientific hypothesis generation. I also needed to translate between different terms and not use the number of citations or the ranking of the tools as an indicator of the quality of the research. The final literature review was, therefore, not just a list of results generated by the tools but a critical reading, classification, comparison and synthesis of the literature.

I would suggest a staged workflow if another student is doing a similar review. First, try Semantic Scholar using multiple combinations of keywords, then filter by relevance, recency and number of citations. Second, utilize ResearchRabbit and Litmaps, utilizing the most effective seed papers to investigate backwards and forwards citations. Third, compare the selected papers using Elicit with respect to objectives, methods, findings, limitations, and research gaps. Lastly, compile all the results in a single spreadsheet, eliminate any duplicate results, check original papers, and categorize them under themes like single-LLM generation, multi-agent debate, evaluation criteria, scientific validation, and research limitations. This is a good combination to get wide coverage but still retain human judgment needed for a credible literature review.

14. Part K — Recommended Workflow

Propose your own workflow after completing the experiment.

Stage	Tool(s) Recommended	Reason
Topic exploration	Semantic Scholar + Elicit	Semantic Scholar helps explore broad keywords and related concepts, while Elicit helps refine the research question and identify major themes.
Initial literature discovery	Semantic Scholar	It provides broad coverage and quickly identifies relevant papers using keyword searches. Sorting by relevance helps locate the closest studies.
Foundational/seminal literature	ResearchRabbit + Litmaps	Both tools are useful for tracing citation networks, finding highly connected papers, and identifying older influential studies.
Recent literature	Semantic Scholar	Its recency-based sorting makes it effective for finding newly published papers and emerging research directions.
Related literature expansion	ResearchRabbit + Litmaps	Citation maps help discover backward and forward references, related authors, and

		connected research that keyword searches may miss.
Literature comparison/synthesis	Elicit	Elicit organizes papers and supports comparison of objectives, methods, findings, limitations, and conclusions.
Research-gap identification	Elicit + manual paper reading	Elicit helps identify repeated limitations and unanswered questions, but human judgement is necessary to determine whether a gap is meaningful and genuine.
Final literature-review writing	Elicit + spreadsheet/word processor + original papers	Elicit provides structured notes, while a spreadsheet helps organize themes and remove duplicates. Original papers must be checked before writing the final review.

15. Required Submission

Submit the following as one assignment package:

- Final literature review/research paper.
- Complete list of papers used in the final review.
- Exact prompts/search queries used with ResearchRabbit, Litmaps, Semantic Scholar and Elicit.
- Comparative analysis of the four tools.
- Completed tool-rating table.
- Quantitative/qualitative observations on the literature discovered by each tool.
- Student's critical remarks.
- Recommended literature-review workflow.
- Screenshots or exported outputs only where they provide useful evidence of tool use.

The central submission is the literature review and the comparative evaluation. Screenshots are supporting evidence, not the primary deliverable.

16. Submission Format

Component	Format
Literature review paper	PDF
Prompts/search queries	Included in PDF and/or separate DOCX/TXT
Tool comparison	Included in PDF
Literature list	Excel/Google Sheet or appendix
Tool screenshots/outputs	PNG/PDF, where required

Suggested file name: LiteratureReview_TopicID_RollNo_Name.pdf

17. Evaluation Rubric

Component	Marks
Quality and scope of literature review	30

Thematic synthesis and critical discussion	20
Use of four research tools	10
Comparative analysis of tools	20
Documentation of prompts/search queries	5
Research-gap and future-direction analysis	10
Student critical remarks and recommended workflow	5
TOTAL	100

18. Final Checklist

- Assigned topic used consistently.
- All four tools used: ResearchRabbit, Litmaps, Semantic Scholar and Elicit.
- Literature review generated and refined using the collected literature.
- Review is thematic and analytical rather than a sequence of paper summaries.
- At least 20 scholarly papers used, unless the instructor specifies otherwise.
- Exact prompts/search queries preserved.
- Contribution of each tool documented.
- Four tools compared using the specified criteria.
- Student's own critical remarks included.
- Recommended workflow provided.
- Final literature review submitted.

19. Student Declaration

I confirm that I used the four specified research tools to explore and analyse literature for my assigned topic, preserved the prompts/search queries used, and prepared the submitted literature review through my own critical evaluation and synthesis of the literature. I understand that AI-assisted research tools support academic work but do not replace independent scholarly judgement.

Field	Response
Student Name	Soumik Bhattacharyya
Roll Number	MDE2026005
Topic ID	T-G11
Date	11/09/2026
Signature	<i>Soumik Bhattacharyya</i>